Vincent Souveton, PhD student

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Education

2021 – 2024 **Ph.D., Applied Mathematics, Université Clermont Auvergne.** The goal of my PhD is to study and develop algorithms to extract meaningful information from astronomical surveys for characterizing the large scale structure of the Universe. I am both interested in Machine Learning techniques and non-reversible Monte Carlo methods.
Thesis title: *Mathematical aspects in statistical inference of initial cosmological parameters through forward modeling.*

2019 – 2021 **M.Sc., Mathematics, UCA.** Various courses about both fundamental and applied Mathematics. Specialization in Partial Differential Equations during the last year.
1st year thesis title: *Holomorphic functions on the disk and Aleksandrov-Clark measures.*
2nd year thesis title: *Mathematical aspects in statistical inference of initial cosmological parameters through forward modeling.*

2017 – 2018 **Ensai Rennes.** National school for Statistics and data analysis. I chose to leave after one year and a half to focus on a research-oriented education.

2015 – 2017 **Classes préparatoires MPSI/MP*, Lycée Blaise Pascal, Clermont-Fd.** Preparatory years for nationwide competitive examination to the French schools of engineering.

2015 **Baccalauréat scientifique, Lycée Jeanne d'Arc, Clermont-Fd.** With very high honours.

Employment / Community life / Responsabilities

2021 – 2024 **Teaching assistant, UCA.** As part of my PhD, I have been giving tutorials to first year science students for two different courses.
Mathématiques S2: *asymptotic analysis and Taylor expansion, vector spaces, linear applications, sequences.*
Outils Mathématiques 2: *logic and reasoning, functions of multiple variables, ordinary differential equations.*

Responsabilities as a PhD student, UCA. During my PhD, I was elected a PhD student representative at the Doctoral School (May 2022 - November 2023). I have also been involved in the deployment of actions for fighting against psychosocial hazard as well as sexist and sexual violence. Finally, I have been in charge of the PhD students seminar organization.

Jan. - June 2021 **Master's degree internship, UCA.** Interdisciplinary research internship between applied Mathematics and Cosmology. My work consisted in a bibliography search and I produced theoretical results regarding the convexity analysis of a sampling problem.
Thesis title: *Mathematical aspects in statistical inference of initial cosmological parameters through forward modeling.*

2020 – 2021 **Tutoring, UCA.** The job consisted in helping first year science students with their maths homework and guide them through efficient preparation for the exams.

Employment / Community life / Responsabilities (continued)

July 2018 — **Volunteer, Archelon.** This Greek NGO aims at the protection of sea turtles. I was in charge of patrolling along the beaches, making public awareness and participating in the daily life of our international camp based in Matala (Crete).

2017 – 2018 — **Administrator, Ensai Junior Consultant.** EJC is the junior enterprise of Ensai. It provides statistical analysis made by students from the school for clients both in the private and the public sector. As a quality rep, my job consisted in following the different projects, making sure that the legal framework was respected.

Research Publications

Preprint

1. V. Souveton, A. Guillin, J. Jasche, G. Lavaux, and M. Michel, *Fixed-kinetic neural hamiltonian flows for enhanced interpretability and reduced complexity*, 2023. arXiv: `2302.01955 [cs.LG]`.

Talks

11/28/2023 — *Sampling with Neural Hamiltonian Flows.* Flashtalk at Institut d'Astrophysique de Paris during the "Debating the potential of Machine Learning in atronomical surveys" conference.

11/22/2023 — *Introduction to Geometric Deep Learning.* Presented at the PhD students seminar in Laboratoire de Mathématiques Blaise Pascal (Clermont-Ferrand).

11/07/2023 — *Algorithms for inferring the initial conditions of the Universe.* Talk as part of an interdisciplinary public colloquium called "Le Puy de la Recherche" (Clermont-Ferrand).

09/20/2023 — *Sampling with Neural Hamiltonian Flows.* Talk during the workshop "Probabilistic sampling for physics: finding needles in a field of high-dimensional haystacks" at Institut Pascal (Orsay).

12/15/2022 — *Sampling with Hamiltonian Normalizing Flows.* Presented at the Simatlab seminar as part of a scientific collaboration between Université Clermont Auvergne and Michelin.

11/25/2021 — *Inferring the initial conditions of the Universe.* Presented at the Cosmology group seminar at the Oskar Klein Center (Stockholm).

Skills

Languages	French (native speaker), English (full professional capacity) and Spanish (basis).
Coding	Python/PyTorch, Julia, LaTeX.
Web Dev	Basic knowledge of HTML and MarkDown.
Misc.	Driver's license.